

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

Make up Examinations - May - June 2023

21BT304 - MOLECULAR BIOLOGY AND GENETIC ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Draw diagrams when necessary.

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) With illustration explain the role of nucleosomes in packing the DNA in the chromosome.	10	L*1	1	1
	b) How is DNA as genetic material proved? Justify with suitable experimental example.	10	L1	2	1
2.	a) Write short notes on: i) Karyotyping ii) Classification of chromosomes.	5+5	L1	1	1
	b) Discuss in detail Nucleotide excision repair.	10	L1	2	1
3.	a) What is DNA damage? How is it different from mutation? Give a detailed account on DNA damage.	10	L1	1	1
	b) Explain about the prokaryotic DNA replication with appropriate illustration.	10	L1	2	1
Unit – II					
4.	a) Discuss post-transcriptional modification.	10	L1	3	1
	b) List out the differences between prokaryotic and eukaryotic protein synthesis.	10	L1	4	1
5.	a) Give a detailed account of transcription in prokaryotes.	10	L1	3	1
	b) Write short note on: i) Genetic Code, ii) Post-translational modifications.	5+5	L1	4	1
6.	a) Delineate the mechanism of regulation of gene expressions in prokaryotes with appropriate example.	10	L1	3	1
	b) With appropriate illustration, give a detailed account of the regulation of eukaryotic gene expression.	10	L1	4	1
Unit – III					
7.	a) Define restriction endonucleases. With an example, give the classification approaches of Restriction endonucleases.	10	L1	5	1
	b) pBR322 is an ideal vector for gene cloning experiments. Justify the statement.	10	L1	5	1
8.	a) Discuss the role of Ti plasmids in the generation of transgenic plants.	10	L1	5	2
	b) Write short note on: i) Cosmids, ii) Phages.	5+5	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
